

# MAKOL CHUOL

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## EDUCATION

**Macalester College**—B.A. Computer Science, Minors: Mathematics, Educational Studies Expected May 2027

**GPA:** 3.72/4.00 (Spring 2026 GPA: 4.0)

**Honors:** Kofi Annan Scholar, Dean's List, Fleming First-Generation Scholar, Full-Ride Scholarship

**Relevant Coursework:** Algorithms, Computer Systems, Data Structures, Discrete Math, CodePath Cybersecurity

## SKILLS & CERTIFICATIONS

**Cloud & Security:** AWS (IAM, ECS, S3, Lambda, CloudWatch, KMS), Terraform, GCP, Azure, OAuth2/OIDC, JWT, SIEM/ELK Stack, NIST Cybersecurity Framework, Wireshark

**Languages:** Python, Java, C/C++, JavaScript/TypeScript, Bash, Swift

**Tools & DevOps:** Docker, Kubernetes, Git, REST APIs, Linux (CLI), OOP, Keycloak, Temporal, Gunicorn, Locust, Jira

**Certifications:** CodePath Cybersecurity Certificate (CYB 102), Cisco Cybersecurity Essentials

## EXPERIENCE

**Software Development Intern** | MacroFab | Houston, TX Jun 2026–Aug 2026

- Delivered 100+ pull requests and resolved 75 JIRA issues over a 10-week internship spanning authentication, infrastructure, and security workstreams, the highest output on the team outside of senior engineers
- Migrated 22 Python microservices to Gunicorn --preload, implementing post\_fork connection-pool resets for DB/AWS clients; validated stability and throughput via Locust load testing and unit tests
- Upgraded Keycloak (24 → 26.2) on a highly-available ECS deployment (rolling-redeploy tested) and validated the end-to-end Temporal OAuth2/JWT authentication flow now approved for production rollout
- Built a reusable GitHub composite action for Python dead code detection (Vulture) with machine-readable JSON output for CI pipelines
- Independently transitioned the automated PR-review agent from OpenAI to Anthropic's Pi; its output was credited with catching a bug that would have caused a production incident
- Authored ADRs for Temporal service-to-service authentication, evaluating mTLS, OAuth2/JWT bearer tokens, and reverse-proxy options; drove adoption of Keycloak-issued JWTs via OAuth2 client-credentials flow

**Cybersecurity Developer Intern** | CrossRealms International | Chicago, IL Feb 2026–May 2026

- Developed scalable telemetry pipelines to detect anomalous activity across cloud and network environments, supporting proactive threat detection and monitoring

**Cybersecurity Engineering Intern** | CrossRealms International | Chicago, IL May 2025–Aug 2025

- Deployed AI-assisted cloud security systems across Azure to detect, classify human, bot, and automated cyberattacks
- Automated honeypot deployment across 5 data centers via Terraform and GitHub Actions, cutting setup to < 10 minutes
- Analyzed 278K+ real-world attacks to generate threat intelligence supporting detection and response strategies

**Preceptor, Software Design & Development (COMP 225)** | Macalester College | St. Paul, MN Aug 2025–Dec 2025

- Mentored 35+ students building secure full-stack applications (React, Node.js, TypeScript, Swift); reviewed code for secure coding practices and maintainability

## ACADEMIC PROJECTS

- CodePath Cybersecurity Project—audited Linux systems and simulated attacks via Bash scripting and system call tracing; applied NIST Cybersecurity Framework and OOP concepts

## LEADERSHIP & ADDITIONAL

- Voices of Amani (Davis Projects for Peace, \$10,000 grant)—designed and led a peace education training program at Kakuma Refugee Camp, equipping 12 teachers as "Peace Champions" reaching 300+ students in conflict resolution and cross-cultural understanding
- Macathon 2024—1st Place: led a 5-person team building an energy-generating soccer ball for off-grid refugee schools; pitched on technical feasibility and social impact
- Kakuma Tech Hub—Founding Lead & Curriculum Developer: designed cloud/AI-based digital learning programs and mentored 80+ students on IT fundamentals in a resource-limited setting